STRAIGHTENING LAWS IN ALGEBRAIC COMPLEXITY

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ABSTRACT.

CONTENTS

1. Introduction

Many problems in combinatorics, geometry, and computer science can be expressed in terms of the computation of some polynomials. For example, finding a perfect matching in a graph is equivalent to determining if it's *Tutte Matrix* has determinant identically zero [Tut47]. With this, we see that it is important to study the complexity of computing polynomials in general, and specifically computing special polynomials such as the determinant or permanent polynomials. Algebraic complexity theory is the study of the complexity of such computations.

Studying the optimality of the number of operations to compute a given polynomial is one of the core problems of algebraic complexity. This is known as the *minimal circuit complexity* problem. Arguably, the first work in this direction was a proof of optimality of *Horner's Rule* (Example 2.4) where Ostrowski showed that one cannot compute a generic degree d univariate polynomial using fewer than d additions. Strassen showed that one can multiply matrices faster than the usual algorithm taught in classes [Str69]. Csanky give fast parallel algorithms for various problems in linear algebra, in particular for computing the determinant [Csa76]. See the survey of Saptharishi for further details of progress in this direction [Sap21] Another important problem in algebraic complexity is the problem of deciding if a given algebraic identity is identically zero. This is known as the *polynomial identity testing* problem. A fast randomized algorithm for this problem over arbitrary fields was independently discovered by Schwartz and Zippel [Sch80; Zip79]. A version for finite fields was given earlier by Ore in 1922 [Ore22]. Derandomizing algorithms for polynomial identity testing is deeply connected to proving lower bounds on minimal circuit size [NW94; KI04]. For further progress on derandomizing polynomial identity testing, the survey of Dutta and Lysikov is a great reference [DL25].

In computational complexity, one tries to characterize the complexity of a given problem and determine a *complexity class* that it lives in. For example, the class P is the set of decision problems, yes and no problems, whose solution can be computed in polynomial time. A closely related class is NP, the set of decision problems whose solution can be *verified* in polynomial time. The question of separating P and NP is the largest open problem in computer science. To do this, one strategy is to find so called *complete* problems in the class NP, and show that these problems are not contained in P. Complete problems in a class such as NP have the property that every problem instance in NP can be efficiently transformed into an instance of the complete problem. For example, the question of finding a satisfying assignment for a boolean formula is NP-complete. Thus, every problem in NP can be efficiently transformed into a boolean formula which is satisfiable if and only if the original problem has a yes solution. This process is known as *reducing* to a complete problem. It is believed that boolean satisfiability is not in ¶.

Valient in two seminal works defined algebraic analogues of complexity classes, analogues of complete problems for these complexity classes, and a notion of reduction [Val79; Val81]. Valient's algebraic analogue of P, known as VP, can be described as n-variate polynomials with poly(n)-degree which are efficiently computable. The analogue of NP, known as VNP, can be described as n-variate polynomials with poly(n)-degree whose *coefficients* are efficiently computable. See Definition 2.13 and Definition 2.16 for formal statements. The complete polynomial for VNP is the $n \times n$ permanent. While the determinant is known to be efficiently computable in many standard algebraic models of computation [Csa76], it is not known if the permanent can be efficiently computed. In fact, it is conjecured that this is not the case. Specifically, say $X = (x_{i,j})$ is a $n \times n$ matrix of variables. Let $m \geq n$ and suppose that A is a matrix of polynomials of degree at most 1 such that $\det_m(A) = \operatorname{perm}_n(X)$. Then we cannot have that $m = \operatorname{poly}(n)$, m must be superpolynomial in n.

1.1. Geometric Complexity Theory. ⟨⟨ Talk about Mulmuley-Sohoni ⟩⟩

1.2. Straightening Laws. One of the most classical examples of standard monomial theory and straightening laws comes from the coordinate ring of the Grassmannian, whose relations are *(quadratic)-Plücker relations*. Many good references on this topic can be found [Ful96; LB18].

Let V be an n-dimensional vector space over a field $\mathbb{F}$. Let $\{e_1, \ldots, e_n\}$ be a basis for V. We will notate by $\Lambda(n,d)$ the set of all ordered tuples $(i_1 < \cdots i_d)$ where each $i_k \in [n]$.

Definition 1.1 (Grassmannian): Let $\operatorname{Gr}(n,d) = \{W \subseteq V \mid \dim_{\mathbb{F}}(W) = d\}$ be the d-dimensional *Grassmannian*.

One can give this the structure of an algebraic variety via the following map.

Definition 1.2 (Plücker Embedding, Plücker Coordinates): For $U \in \operatorname{Gr}(n,d)$, let U have basis $\{u_1, \ldots, u_d\}$. Define a map $\operatorname{Gr}(n,d) \to \mathbb{P}\left(\bigwedge^d V\right)$ given by $U \mapsto [u_1 \wedge \cdots \wedge u_d]$. This map is called the *Plücker embedding*.

The set $\{e_{\bar{i}} = e_{i_1} \wedge \cdots \wedge e_{i_d} \mid \bar{i} \in \Lambda(n,d)\}$ gives a basis for $\bigwedge^d V$. Let $\{p_{\bar{j}}\}_{\bar{j} \in \Lambda(n,d)}$, denote the basis of $\left(\bigwedge^d V\right)^*$ dual to $\{e_{\bar{i}}\}_{\bar{i} \in \Lambda(n,d)}$. Then for each $U \in \operatorname{Gr}(n,d)$, we can represent U via an $n \times d$ matrix of rank d via the basis $\{u_1, \ldots, u_d\}$. Then $p_{\bar{j}}(U)$ is the $d \times d$ determinant of the minor defined by rows $\bar{j} \in \Lambda(n,d)$. The coordinates $p_{\bar{j}}$ are known as *Plücker coordinates*. We will often identify the Plücker coordinate $p_{j_1,\ldots,j_d}$ with the associated determinant.

Definition 1.3 (Plücker Relations): For each $(i_1, \ldots, i_{d-1}) \in \Lambda(n, d-1)$ and $(j_1, \ldots, j_{d+1}) \in \Lambda(n, d+1)$, define the following quadratic polynomial in the Plücker coordinates:

$$\sum_{i=1}^{d+1} p_{i_1,\ldots,i_{d-1},j_i} \cdot p_{j_1,\ldots,\widehat{j_i},\ldots,j_{d+1}}.$$

The ideal $\mathfrak{q}$ of these quadratic polynomials is the ideal of *Plücker relations*.

Note that the ideal $\mathfrak{q}$ of quadratic relations in the polynomial ring $\mathbb{F}[p_{\bar{j}}]$ is a prime ideal [Ful96, §8.4, Proposition 2].

Proposition 1.4 ([LB18, Theorem 12.1.3]): The zero set of $\mathfrak{q}$ is the Grassmannian $\mathrm{Gr}(n,d)$.

Definition 1.5 (Plücker / Bracket Algebra): By Proposition 1.4, the coordinate ring $\mathbb{F}[\mathrm{Gr}(n,d)]$ of $\mathrm{Gr}(n,d)$ is given by

$$\mathbb{F}[p_{\bar{i}} \mid \bar{i} \in \Lambda(n,d)]/\mathfrak{q}.$$

In fact, one can also show that this coordinate ring is the invariant subring $\mathbb{F}[x_{i,j} \mid 1 \le i \le n, 1 \le j \le d]^{\mathrm{SL}_d(\mathbb{F})}$. This algebra is known as the *Plücker* or *bracket algebra*.

The fact that the invariant subring $\mathbb{F}[x_{i,j}]^{\mathrm{SL}_d(\mathbb{F})}$ is generated by the $d \times d$ minors is known as the *First Fundamental Theorem of Invariant Theory*. That the relations between the minors are all generated by the Plücker relations is known as the *Second Fundamental Theorem of Invariant Theory*.

We can represent the product of k minors using a $d \times k$ tableau. One standard reference for this theory is [Stu08].

Definition 1.6 (Tableau, Tableaux Ordering, Standard Monomials): We can order the minors $p_{i_1,\dots,i_d} \prec p_{j_1,\dots,j_d}$ if there is some $1 \le m \le d$ such that $i_k = j_k$ for $1 \le k \le m-1$ and $i_m < j_m$. This gives a total order on the Plücker coordinates $p_{\bar{j}}$ which induces a degree reverse lexicographic monomial ordering on the bracket algebra $\mathbb{F}[p_{\bar{j}}]$. This ordering is known as the *tableaux ordering* as we can express a product of k coordinates $\bar{i}^1 \preceq \cdots \preceq \bar{i}^k$ as a tableau

$$\begin{bmatrix} i_i^1 & \cdots & i_d^1 \\ i_i^2 & \cdots & i_d^2 \\ & \vdots & \\ i_i^k & \cdots & i_d^k \end{bmatrix}.$$

We will say that such a monomial is *standard* if the columns are sorted (recall the rows are already sorted by convention). Otherwise, such a monomial is *nonstandard*.

This *standard monomial* theory is originally due to Hodge [Hod43]. However, to express the product of standard monomials in terms of other standard monomials, we use a *straightening* whose formulation is due to Young [You28].

Definition 1.7 (van der Waerden Syzygies, Straightening Syzygies): For $\overline{\lambda} \in \Lambda(n,d)$, let $\overline{\lambda^*}$ be the *complement* tuple in $\Lambda(n,n-d)$ consisting of the $n-d$ elements which do not appear in $\overline{\lambda}$. The *sign* of $\overline{\lambda}$ and $\overline{\lambda^*}$, denoted by $\mathrm{sgn}(\overline{\lambda},\overline{\lambda^*})$, is the sign of the permutation $\sigma \in \mathfrak{S}_n$ given in one-line notation by $\lambda_1 \cdots \lambda_d \lambda_1^* \cdots \lambda_{n-d}^*$.

For $s \in [d]$, $\overline{\alpha} \in \Lambda(n, s-1)$, $\overline{\beta} \in \Lambda(n, d+1)$, and $\overline{\gamma} \in \Lambda(n, d-s)$, the *van der Waerden syzygy* $\left[\left[\overline{\alpha}\dot{\overline{\beta}}\overline{\gamma}\right]\right]$ is the following quadratic polynomial in $\mathbb{F}[p_{\overline{i}} \mid \overline{i} \in \Lambda(n, d)]$:

$$\left[\left[\overline{\alpha}\dot{\overline{\beta}}\overline{\gamma}\right]\right] := \sum_{\overline{\tau} \in \Lambda(d+1, s)} \operatorname{sgn}(\overline{\tau}, \overline{\tau^*}) \cdot p_{\alpha_1, \ldots, \alpha_s, \beta_{\tau_1^*}, \ldots, \beta_{\tau_{d+1-s}^*}} \cdot p_{\beta_{\tau_1}, \ldots, \beta_{\tau_s}, \gamma_1, \ldots, \gamma_{d-s}}.$$

Such a van der Waerden syzygy is a *straightening syzygy* if $\alpha_{s-1} < \beta_{s+1}$ and $\beta_s < \gamma_s$. Note that if we take $s = 1$, we get the Plücker relations.

Theorem 1.8 ([Stu08, Theorem 3.1.7], [Rai]): The set of straightening syzygies is a Gröbner basis for the ideal of quadratic Plücker relations $\mathfrak{q}$ under the tableaux order. A monomial is standard if and only if it does not lie in the initial ideal $\mathrm{LM}_{\preceq}\, \mathfrak{q}$. Thus, the set of standard monomials forms an $\mathbb{F}$-vector space basis for $\mathbb{F}[\mathrm{Gr}(n, d)]$. If we declare each $p_{\overline{i}}$ to have degree one, then the standard tableau of shape $d \times k$ give a basis for the k-th graded component of $\mathbb{F}[\mathrm{Gr}(n, d)]$.

Example 1.9: **⟨⟨ Example straightening calculations will go here ⟩⟩**

2. Preliminaries on Algebraic Complexity

2.1. Computational Models.

Definition 2.1 (Algebraic Circuits): An *algebraic circuit* over a field $\mathbb{F}$ is a directed acyclic graph whose *input nodes*, nodes of in-degree zero, are labeled either by variables $x_1, \ldots, x_n$ or constants from $\mathbb{F}$. Nodes of in-degree greater than zero are labeled by Σ or Π which are *addition gates* or *multiplication gates* respectively. Each edge or *wire* is also labeled by a constant from $\mathbb{F}$.

Computation is defined inductively node by node. Input nodes compute their label. Every Σ gate computes the sum of their children multiplied by the constants along each wire, and computation at Π gates is defined similarly. Nodes of out-degree zero are *outputs*. We say that an algebraic circuit *computes* $\{f_1(\overline{x}), \ldots, f_k(\overline{x})\}$ if these are the polynomials computed by all the outputs.

The *size* of a circuit is the total number of edges. We will denote by $size(f_1, \ldots, f_k)$ the size of the smallest circuit over $\mathbb{F}$ computing $\{f_1(\overline{x}), \ldots, f_k(\overline{x})\}$. Some sources also define the size to be the number of vertices. These are polynomially related as a directed simple graph with v vertices has $O(v^2)$ edges. It's *depth* is the length of the longest input-to-output path. The *fan-in* and *fan-out* of a node are in-degree and out-degree respectively. The *fan-in* and *fan-out* of the circuit is the maximum fan-in and fan-out respectively over all nodes in the circuit. The *degree* or *syntactic degree* of a circuit is the maximum degree of a polynomial computed by any of the nodes of the circuit.

Definition 2.2 (Constant Depth Circuits): Without loss of generality, we can assume that a circuit is made up of alternating layers of Σ and Π gates. We can use this notation to describe a restriction on the depth and types of gates appearing on each layer. For example

- A $\Sigma\Pi$ circuit computes a polynomial via it's monomial or *sparse* representation.
- A $\Pi\Sigma$ circuit computes a polynomial as a product of degree ≤ 1 factors.

Definition 2.3 (Minimal Circuit Size Problem): Given an n-variate polynomial $f(x_1, \ldots, x_n)$, find an algebraic circuit of minimal size computing $f(x_1, \ldots, x_n)$.

Example 2.4: Let $f(x) = \sum_{i=0}^{d} a_i x^i \in \mathbb{F}[x]$ be a univariate polynomial.

Then *Horner's Rule* gives a recurrence for efficiently computing $f(x)$:

$$p_0(x) = a_0, \qquad p_k(x) = a_{d-k} + x \cdot p_{k-1}(x).$$

This recurrence can be formulated as an algebraic circuit of size $O(d)$ and depth $O(d)$. In fact, this uses exactly n additions and n multiplications, which is optimal. The proof of optimality of the number of additions for computing a generic univariate polynomial $f(x)$ in this manner was given by [Ost54]. The corresponding proof of optimality for multiplications was given by [Pan66].

Definition 2.5 (Algebraic Formulas): An *algebraic formula* over a field $\mathbb{F}$ is an algebraic circuit whose underlying graph is a tree.

Definition 2.6 (Oracle Circuits and Formulas): For a fixed polynomial $f(\overline{x}) \in \mathbb{F}[\overline{x}]$, an f*-oracle circuit* is an algebraic circuit defined in the same manner as Definition 2.1, however instead of just Σ and Π gates we now additionally allow $f(\overline{x})$ gates. Note that such gates may have some inputs set as wires and some inputs set as fixed constants from $\mathbb{F}$. We also define an f*-oracle formula* as an f-oracle circuit whose underlying graph is a tree.

Definition 2.7 (Algebraic Branching Programs): An *algebraic branching program* (*ABP*) over a field $\mathbb{F}$ is a directed acyclic graph A such that:

- The nodes V of A are partitioned into non-empty subsets $V = V_0 \sqcup \cdots \sqcup V_d$ for some integer $d \geq 0$ and we call V_i the i-th layer. Layer V_0 has exactly one node called the *source* s and layer V_d has exactly one node called the *sink* t.
- Each edge in A goes from layer V_{i-1} to V_i for some $i \in [d]$.
- Each edge e in A is labeled with a polynomial $\gamma_e \in \mathbb{F}[\overline{x}]$ of degree at most one.

For a path $\overline{e} = (e_1, \ldots, e_d)$ from s to t, where each edge e_i goes from layer $i-1$ to layer i, define the polynomial $\gamma_{\overline{e}} := \prod_{i=1}^{d} \gamma_{e_i}$. Then A computes the polynomial $\sum_{\substack{\text{path } \overline{e} \\ \text{from } s \text{ to } t}} \gamma_{\overline{e}}$.

The *size* of an ABP A is the number of edges in A. Some sources also define the size to be the number of vertices. These are polynomially related as a directed simple graph with v vertices has has

$O(v^2)$ edges. The *width* of A is the maximum number of vertices in any layer of A. The *depth* of A is the number of layers.

Remark 2.8: Definition 2.7 is really the definition of a *layered* ABP, where each edge only goes from one layer to the next without skipping any layers. However, one can convert a non-layered ABP to a layered ABP with only a quadratic blowup in size.

Definition 2.9 (Determinantal Complexity): For a polynomial $f(\overline{x})$ over a field $\mathbb{F}$, the *determinantal complexity* of $f(\overline{x})$, denoted $\mathrm{dc}(f)$, is the smallest m such that there exists an $m \times m$ matrix M with entries in $\mathbb{F}[\overline{x}]_{\leq 1}$ such that $\det_m(M) = f(\overline{x})$.

Lemma 2.10 ([Val79; Bür24]): Suppose that $f(\overline{x})$ can be computed by an algebraic branching program of size m. Then $\mathrm{dc}(f) \leq m$.

In fact, we can say much more about the connection between the complexity of branching programs and determinants:

Lemma 2.11 ([AF22, Lemma 3.6], [Val79, Theorem 1]): Let $g(\overline{y}) \in \mathbb{F}[\overline{y}]$ and suppose g can be computed by an algebraic branching program on r vertices. Then there is an $r \times r$ matrix A over $\mathbb{F}[\overline{y}]$ whose entries are polynomials in $\overline{y}$ of degree at most 1 such that

- $\det_r(A) = 1 + g(\overline{y})$, and
- for all $1 \leq k < r$, $\det_k(A_{[k],[k]}) = 1$.

Algebraic branching programs are closed under homogenization, as stated in the following lemma.

Lemma 2.12 ([AF22, Lemma 3.7]): Let $g(\overline{y}) \in \mathbb{F}[\overline{y}]$ and suppose g can be computed by an algebraic branching program on m vertices. Let z be a new indeterminate. Then there is a homogeneous polynomial $\widehat{g}(\overline{y}, z)$ such that $g = \widehat{g}(\overline{y}, 1)$ and $\widehat{g}$ can be computed by an algebraic branching program on m vertices.

2.2. Complexity Classes.

Definition 2.13 (VP): The class VP is the set of polynomials $f(x_1, \ldots, x_n)$ of degree $\mathrm{poly}(n)$ which are computable by an algebraic circuit of size $\mathrm{poly}(n)$.

Definition 2.14 (VF): The class VF is the set of polynomials $f(x_1, \ldots, x_n)$ of degree $\mathrm{poly}(n)$ which are computable by an algebraic formula of size $\mathrm{poly}(n)$.

Definition 2.15 (VBP): The class VBP is the set of polynomials $f(x_1, \ldots, x_n)$ of degree $\mathrm{poly}(n)$ which are computable by an algebraic branching program of size $\mathrm{poly}(n)$.

Definition 2.16 (VNP): The class VNP is the set of polynomials $f(x_1,\dots,x_n)$ such that there is some $g(x_1,\dots,x_n,y_1,\dots,y_m)$ in VP, where $m = \mathrm{poly}(n)$, such that

$$f(x_1,\dots,x_n) = \sum_{\overline{a}\in\{0,1\}^m} g(x_1,\dots,x_n,a_1,\dots,a_m).$$

Valient's Criterion gives a useful characterization for what it means for $f(\overline{x})$ to be in VNP.

Theorem 2.17 ([Val79]): Let $f = \sum_{\overline{e}} f_{\overline{e}} x_1^{e_1}\cdots x_n^{e_n}$. Consider a function ϕ which take as input $\overline{e}$ and computes the coefficient $f_{\overline{e}}$. If ϕ is computable in polynomial time*, then $f \in$ VNP.

We give some discussion on the restriction of poly(n) degree for VP:

Remark 2.18: Note that via repeated squaring, one can compute polynomials of exponential degree with small circuits, even with restrictions on the fan-in of gates. However, most "interesting" polynomials such as the determinant and permanent fall into this low-degree regime. Originally, Valiant was studying formulas, not circuits. One can easily see from Definition 2.5 that a formula of depth Δ must compute a polynomial of degree $O(\Delta)$ if we restrict to constant fan-in. See [Sap21] and [Gro] for more discussion.

2.3. **Basic Structural Results.**

Lemma 2.19: **⟨⟨ Depth reduction, conversions from circuits to formulas / ABPs and vice versa. ⟩⟩**

Proposition 2.20: **⟨⟨ Inclusions ⟩⟩** VF $\subseteq$ VBP $\subseteq VP$

Lemma 2.21: **⟨⟨ Elimination of Division ⟩⟩**

Definition 2.22: For $f \in \mathbb{F}[\overline{x}, t]$ and an integer $i \in \mathbb{Z}_{\geq 0}$, denote by $\mathrm{coeff}_{t^i}(f)$ the coefficient of t^i in f when f is viewed as a univariate polynomial in t over the ring $\mathbb{F}[\overline{x}]$.

Lemma 2.23 ([DG25, Lemma 2.29]): Let $g \in \mathbb{F}[\overline{x}, t]$, and let $f \in \mathbb{F}[\overline{x}, t]$ be a degree-d polynomial such that for each $\alpha \in \mathbb{F}$, the polynomial $f(\overline{x}, \alpha)$ is computed by a g-oracle circuit C_α of size at most s and depth at most Δ. Assume that $|\mathbb{F}| \geq d+1$. Then, for every $0 \leq i \leq d$, there exists a g-oracle circuit D_i of size $O(ds)$ and depth $\Delta + 1$ that computes $\mathrm{coeff}_{t^i}(f)$, with its top gate being an addition gate and bottom Δ layers consisting of $d+1$ copies of g-oracle circuits of the form C_α. Moreover, if the top gate of each C_α is an addition gate, then the depth bound can be improved to Δ.

Proof: By definition,

$$f = \sum_{i=0}^{d} \mathrm{coeff}_{t^i}(f) t^i.$$

*Formally, we mean that ϕ is in the complexity class P / poly. However, saying "polynomial time" is sufficient for this purpose.

Let $\alpha_1, \dots, \alpha_{d+1}$ be $d+1$ distinct elements in $\mathbb{F}$. Then we have that

$$\begin{pmatrix} \alpha_1^d & \alpha_1^{d-1} & \cdots & \alpha_1 & 1 \\ \alpha_2^d & \alpha_2^{d-1} & \cdots & \alpha_2 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \alpha_d^d & \alpha_d^{d-1} & \cdots & \alpha_d & 1 \\ \alpha_{d+1}^d & \alpha_{d+1}^{d-1} & \cdots & \alpha_{d+1} & 1 \end{pmatrix} \begin{pmatrix} \mathrm{coeff}_{t^d}(f) \\ \mathrm{coeff}_{t^{d-1}}(f) \\ \vdots \\ \mathrm{coeff}_{t}(f) \\ \mathrm{coeff}_{1}(f) \end{pmatrix} = \begin{pmatrix} f(\overline{x}, \alpha_1) \\ f(\overline{x}, \alpha_2) \\ \vdots \\ f(\overline{x}, \alpha_d) \\ f(\overline{x}, \alpha_{d+1}) \end{pmatrix}$$

The $(d+1) \times (d+1)$ matrix $(\alpha_i^{d+1-j}) \in \mathbb{F}^{(d+1)\times(d+1)}$ is a Vandermonde matrix. It's determinant is $\prod_{i<j}(\alpha_j - \alpha_j)$. As all the α_i are distinct, this determinant is nonzero so that the matrix is invertible. Thus, there exists a $(d+1) \times (d+1)$ matrix $(z_{i,j}) \in \mathbb{F}^{(d+1)\times(d+1)}$ such that

$$\begin{pmatrix} \mathrm{coeff}_{t^d}(f) \\ \mathrm{coeff}_{t^{d-1}}(f) \\ \vdots \\ \mathrm{coeff}_{t}(f) \\ \mathrm{coeff}_{1}(f) \end{pmatrix} = \begin{pmatrix} z_{1,1} & z_{1,2} & \cdots & z_{1,d} & z_{1,d+1} \\ z_{2,1} & z_{2,2} & \cdots & z_{2,d} & z_{2,d+1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ z_{d,1} & z_{d,2} & \cdots & z_{d,d} & z_{d,d+1} \\ z_{d+1,1} & z_{d+1,2} & \cdots & z_{d+1,d} & z_{d+1,d+1} \end{pmatrix} \begin{pmatrix} f(\overline{x}, \alpha_1) \\ f(\overline{x}, \alpha_2) \\ \vdots \\ f(\overline{x}, \alpha_d) \\ f(\overline{x}, \alpha_{d+1}) \end{pmatrix}$$

Fix $0 \le i \le d$. We now have an explicit expression for $\mathrm{coeff}_{t^i}(f)$:

$$\mathrm{coeff}_{t^i}(f) = \sum_{j=1}^{d+1} z_{d+1-i,j} f(\overline{x}, \alpha_j) \tag{1}$$

With Equation (1), we can now describe the circuit for $\mathrm{coeff}_{t^i}(f)$. The scalars $\alpha_1, \dots, \alpha_{d+1}$ are fixed a priori, and hence so are the $z_{i,j}$ for $i, j \in [d+1]$. Recall that scalar multiplications along wires are free. For each $j \in [d+1]$, computing $f(\overline{x}, \alpha_j)$ requires a single g-oracle circuit C_{α_j} of size at most s and depth Δ. Hence, evaluating all $d+1$ such terms requires total size $O(ds)$. The top-level summation uses one addition gate and $d+1$ scalar multiplications by the $z_{d+1-i,j}$ along $d+1$ wires. Therefore, Equation (1) yields a g-oracle circuit D_i of size $O(ds + d) = O(ds)$ and depth $\Delta + 1$ computing $\mathrm{coeff}_{t^i}(f)$. Moreover, if the top gate of each C_{α_j} is an addition gate, then the top two layers of D_i consist entirely of additions and can be merged to reduce the depth bound to Δ. □

Remark 2.24: The condition $|\mathbb{F}| \ge d+1$ in Lemma 2.23 is required for interpolation. We note that if g is given not as a black box but as a white-box circuit, then its homogeneous components can be extracted in a gate-by-gate fashion [Str73], which requires no lower bound on $|\mathbb{F}|$.

Lemma 2.25 ([KS01, Lemma 4]): Let C be any collection of distinct linear forms in variables $z_1, \dots, z_\ell$ with coefficients in the range $\{0, \dots, K\}$ for some integer $K \in \mathbb{Z}_{\ge 0}$. Let $\varepsilon > 0$. Let $z_1, \dots, z_\ell$ be independently and uniformly chosen from $\{0, \dots, M\}$ at random, where $M \ge K\ell/\varepsilon$. Then, with probability at least $1 - \varepsilon$, there is a unique form in C of minimum value at $z_1, \dots, z_\ell$.

A simple rephrasing of Lemma 2.25 allows us to isolate a term of a multivariate polynomial $f(y_1,\dots,y_\ell)$ under a random monomial substitution $y_i \mapsto w^{z_i}$:

Corollary 2.26 ([DG25, Corollary 2.27]): Let $K \in \mathbb{Z}_{\geq 0}$. Let A be any ring, and let $f(y_1,\dots,y_\ell) \in A[\overline{y}]$ be a polynomial whose individual degree in each variable is at most K, i.e., $\deg_{y_i}(f) \leq K$ for all $i \in [\ell]$. Fix $\varepsilon > 0$. Choose $z_1,\dots,z_\ell$ independently and uniformly at random from $\{0,\dots,M\}$, where $M \geq K\ell/\varepsilon$. Let w be a new indeterminate, and define $\phi : A[y_1,\dots,y_\ell] \to A[w]$ to be the ring homomorphism sending $y_i \mapsto w^{z_i}$ for each $i \in [\ell]$. Then, with probability at least $1-\varepsilon$, there exists a unique monomial $\mathfrak{m} = y_1^{e_1}\cdots y_\ell^{e_\ell}$ among all monomials of f such that $\phi(\mathfrak{m}) = w^{e_1 z_1+\cdots+e_\ell z_\ell}$ attains the minimum degree in w.

The advantage of Corollary 2.26 is when we have a multivariate polynomial in which there are multiple terms of the same total degree. In this situation, suppose that we want a circuit for a single term. Then Lemma 2.23 is insufficient for this purpose as it separates terms by the degree in a single variable. Dealing with multiple variables by repeatedly applying Lemma 2.23 would yield an exponential blowup in the total circuit size. A similar idea would be to apply a *Kronecker* substitution:

Lemma 2.27 ([Kro82]): Write $g(\overline{x}) = \sum_{\overline{e}} \overline{x}^{\overline{e}}$. Suppose that $\deg(g) < d$. Then the *Kroncker substitution*

$$x_i \mapsto w^{d^i}$$

maps distinct monomials to distinct monomials.

One could apply the Kronecker map to our multivariate polynomial and then apply Lemma 2.23 since all terms would have distinct total degree. However, the total degree in w, the new variable, would be exponential so that the resulting circuit would have large size. Here, Corollary 2.26 argues there is an alternative assignment of weights maps $x_i \mapsto w^{z_i}$ rather than $x_i \mapsto w^{d^i}$ such that there is a unique term of lowest total degree in w. Thus, if we can argue that *any* of our terms of the same total degree suffice, we can apply Corollary 2.26 followed by Lemma 2.23 to get a small circuit computing a single term. To the best of our knowledge, this is the first usage of the Isolation Lemma of [MVV87; KS01] in algebraic complexity for the purposes of isolating terms in a polynomial.

2.4. **Border Complexity.**

Definition 2.28 (Border Computation): We say that $g \in \mathbb{F}[x_1,\dots,x_n]$ is *border computed* by a circuit C over $\mathbb{F}(\varepsilon)$ if C computes some polynomial $h \in \mathbb{F}(\varepsilon)[x_1,\dots,x_n]$, and $h-g \in \varepsilon\cdot\mathbb{F}[[\varepsilon]][x_1,\dots,x_n]$, where $\mathbb{F}[[\varepsilon]]$ denotes the ring of formal power series in ε. We will denote by $\overline{\mathrm{size}}(f)$ the size of the smallest circuit over $\mathbb{F}(\varepsilon)$ border computing $f(\overline{x})$.

In the settings where $\mathbb{F}[\varepsilon][x_1,\dots,x_n]$ can be equipped with the Euclidean topology in a natural way, such as $\mathbb{F} = \mathbb{C}$ or $\mathbb{R}$, this would imply that $h \to g$ as $\varepsilon \mapsto 0$. However, this does not mean that we could just substitute ε by zero in C and obtain a circuit exactly computing g since C may use scalars in $\mathbb{F}(\varepsilon)$, such as $1/\varepsilon$, that are not well-defined at $\varepsilon = 0$.

Example 2.29 ([DL25]): In order to interpolate the coefficients of $f(\overline{x}, t) = \sum_{i=0}^{d} \mathrm{coeff}_{t^i}(f) \cdot t^i$ over $\mathbb{F}$ as in Lemma 2.23, one needs that $|\mathbb{F}| \geq d+1$. However, we can border compute $\mathrm{coeff}_{t^r}(f)$ if $|\mathbb{F}| \geq r+1$.

Suppose $\mathbb{F}$ has distinct elements $\{\alpha_0, \dots, \alpha_r\}$. Then, there exists elements $\beta_0, \dots, \beta_r \in \mathbb{F}$ such that

$$\mathrm{coeff}_{t^r}(f) = \lim_{\varepsilon \to 0} \frac{1}{\varepsilon^r} \left(\sum_{i=0}^{r} \beta_i \cdot f(\overline{x}, \varepsilon \alpha_i) \right).$$

Indeed let $g(\overline{x}, t) = \sum_{i=0}^{r} \mathrm{coeff}_{t^i}(f) \cdot t^i$ and $h(\overline{x}, t) = f(\overline{c}, t) - g(\overline{x}, t)$. Then standard interpolation on $\varepsilon \alpha_i$ gives us that there exists field constants $\beta_0, \dots, \beta_r$ such that $\varepsilon^r \mathrm{coeff}_{t^r}(f) = \sum_{i=0}^{r} \beta_i \cdot g(\varepsilon \alpha_i)$. Thus, we have that

$$\begin{aligned}
&\lim_{\varepsilon \to 0} \frac{1}{\varepsilon^r} \left(\sum_{i=0}^{r} \beta_i \cdot f(\overline{x}, \varepsilon \alpha_i) \right) \\
&= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon^r} \left(\sum_{i=0}^{r} \beta_i \cdot g(\overline{x}, \varepsilon \alpha_i) \right) + \lim_{\varepsilon \to 0} \frac{1}{\varepsilon^r} \left(\sum_{i=0}^{r} \beta_i \cdot h(\overline{x}, \varepsilon \alpha_i) \right) \\
&= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon^r} \left(\sum_{i=0}^{r} \beta_i \cdot g(\overline{x}, \varepsilon \alpha_i) \right) + 0 = \mathrm{coeff}_{t^r}(f).
\end{aligned}$$

Example 2.30: **⟨⟨ Bordering Waring Rank ⟩⟩**

⟨⟨ Define border complexity classes ⟩⟩

Example 2.31: **⟨⟨ Add some of Pranjal's examples of separations ⟩⟩**

3. Preliminaries on Straightening Laws

3.1. **Determinantal Ideals.** Let $\mathbb{F}$ be a field. Let $X = (x_{i,j})_{\substack{1 \leq i \leq n \\ 1 \leq j \leq m}}$ be a $n \times m$ matrix of variables. We will denote by $\mathbb{F}[X]$ the polynomial ring $\mathbb{F}[x_{i,j} \mid 1 \leq i \leq n,\ 1 \leq j \leq m]$. We can natural associate a product of minors of X to a pair of Young tableau where the minors are described by the entries of the Young tableau.

Definition 3.1 (Bitableau, bideterminant): Let $\sigma = (\sigma_1 \geq \cdots \geq \sigma_k)$ be a partition. An (n,m)-*bitableau* $(S \mid T)$ of shape σ is a pair of two Young tableau of shape σ with where the entries of S come from $[n]$ and the entries of T come from $[m]$. When n and m are clear from context, we just call $(S \mid T)$ a *bitableau*. For each $i \in [\widehat{\sigma}_1]$, write the i-th row of S and the i-th row of T as $S_i = (S(i,1), \dots, S(i,\sigma_i))$ and $T_i = (T(i,1), \dots, T(i,\sigma_i))$ respectively. The $\sigma_i \times \sigma_i$ minor of X defined by S_i and T_i is the matrix

$$\left(x_{S(i,a),T(i,b)}\right)_{1\le a,b\le\sigma_i} = \begin{pmatrix} x_{S(i,1),T(i,1)} & x_{S(i,1),T(i,2)} & \cdots & x_{S(i,1),T(i,\sigma_i)} \\ x_{S(i,2),T(i,1)} & x_{S(i,2),T(i,2)} & \cdots & x_{S(i,2),T(i,\sigma_i)} \\ \vdots & \vdots & \ddots & \vdots \\ x_{S(i,\sigma_i),T(i,1)} & x_{S(i,\sigma_i),T(i,2)} & \cdots & x_{S(i,\sigma_i),T(i,\sigma_i)} \end{pmatrix}.$$

Then the associated *bideterminant* $(S \mid T)(X)$ is the product of all such minors over $i \in [\widehat{\sigma}_1]$:

$$(S \mid T)(X) := \prod_{i=1}^{\widehat{\sigma}_1} \det_{\sigma_i}\left(x_{S(i,a),T(i,b)}\right)_{1\le a,b\le\sigma_i}.$$

An $n \times n$ determinant is homogeneous of degree n so that $(S \mid T)(X)$ is a homogeneous polynomial of degree $|\sigma|$.

Example 3.2: If $\sigma = (4,2,1)$ and $(S \mid T)$ is the bitableau

$$(S \mid T) = \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 4 & 5 \\ \hline 3 & 6 \\ \cline{1-2} 4 \\ \cline{1-1} \end{array} \;\middle|\; \begin{array}{|c|c|c|c|} \hline 1 & 3 & 5 & 6 \\ \hline 2 & 7 \\ \cline{1-2} 3 \\ \cline{1-1} \end{array} \right),$$

then the bideterminant $(S \mid T)(X)$ is given by

$$\begin{pmatrix} x_{1,1} & x_{1,3} & x_{1,5} & x_{1,6} \\ x_{2,1} & x_{2,3} & x_{2,5} & x_{2,6} \\ x_{4,1} & x_{4,3} & x_{4,5} & x_{4,6} \\ x_{5,1} & x_{5,3} & x_{5,5} & x_{5,6} \end{pmatrix} \begin{pmatrix} x_{3,2} & x_{3,7} \\ x_{6,2} & x_{6,7} \end{pmatrix} \left(x_{4,3}\right).$$

The following lemma shows that every monomial in $\mathbb{F}[X]$ can be expressed as a bideterminant, and hence the bideterminants span $\mathbb{F}[X]$ as an $\mathbb{F}$-vector space. The proof is immediate from the definition.

Lemma 3.3: A degree d monomial $\prod_{i=1}^d x_{r_i,c_i}$ is given by a bideterminant:

$$\left(\begin{array}{|c|} \hline r_1 \\ \hline r_2 \\ \hline \vdots \\ \hline r_d \\ \hline \end{array} \;\middle|\; \begin{array}{|c|} \hline c_1 \\ \hline c_2 \\ \hline \vdots \\ \hline c_d \\ \hline \end{array} \right)(X) = \prod_{i=1}^d x_{r_i,c_i}.$$

Thus, the bideterminants span $\mathbb{F}[X]$.

While the bideterminants span $\mathbb{F}[X]$, there is a specific subset of bideterminants, the *standard* bideterminants, which form an $\mathbb{F}$-basis of $\mathbb{F}[X]$.

Definition 3.4 (Standard bitableau, standard bideterminant): We call a bitableau $(S \mid T)$ and its associated bideterminant $(S \mid T)(X)$ *standard* if S and T are both conjugate semistandard Young tableaux.

The next theorem is known in the literature as the *straightening law*.

Theorem 3.5 ([DRS74, §8, Theorem 3], [dCEP80, Corollary 2.3]): Let $(S \mid T)(X)$ be a bideterminant of shape σ. Then $(S \mid T)(X)$ can be uniquely expressed as a linear combination

$$(S \mid T)(X) = \sum_{(A|B)} c_{A,B}(A \mid B)(X),$$

where $(A \mid B)(X)$ is a standard bideterminant of shape $\tau \geq_{\text{lex}} \sigma$, and $c_{A,B} \in \mathbb{Z}$ when $\text{char}(\mathbb{F}) = 0$, while $c_{A,B} \in \mathbb{Z}/p\mathbb{Z}$ when $\text{char}(\mathbb{F}) = p > 0$.

The fact that bideterminants are homogeneous polynomials, together with Lemma 3.3 and Theorems 3.5, imply the following corollary.

Corollary 3.6: The standard bideterminants form an $\mathbb{F}$-basis of $\mathbb{F}[X]$. In particular, since bideterminants are homogeneous with respect to total degree, the degree-d component of $\mathbb{F}[X]$ has as basis the standard bideterminants of shape σ such that $|\sigma| = d$.

Let $I^{\text{det}}_{n,m,r}$ be the ideal of $\mathbb{F}[X]$ generated by the $r \times r$ minors of X. Note that this ideal is a $\mathbb{F}$-subspace of $\mathbb{F}[X]$. We have the following crucial statement on the $\mathbb{F}$-basis of this ideal.

Proposition 3.7 ([BC03, Corollary 1.7]): Polynomials $f \in I^{\text{det}}_{n,m,r}$ are supported by standard bideterminants of shape σ such that $\sigma_1 \geq r$; that is, by those whose first row has length at least r.

The notion of the total degree of a polynomial is too coarse for differentiating terms in the expansion of a polynomial $f \in I^{\text{det}}_{n,m,r}$ with respect to the standard basis. Thus, we describe a grading of the polynomial ring $\mathbb{F}[X]$ that is better suited for this purpose.

Definition 3.8: The polynomial ring $\mathbb{F}[X]$ has a natural $(\mathbb{N}^n \oplus \mathbb{N}^m)$-grading. Let $\overline{e}_i \in \mathbb{N}^n$ be the standard basis vector with a 1 in position i and 0's in the other positions. Define $\overline{e}_i \in \mathbb{N}^m$ the same way. Then we assign $x_{i,j}$ to have degree $\overline{e}_i \oplus \overline{e}_j$. We call this assignment the *multidegree*. We say a polynomial $f \in \mathbb{F}[X]$ is *multihomogeneous* of multidegree $(s_1\overline{e}_1 + \cdots + s_n\overline{e_n}) \oplus (t_1\overline{e}_1 + \cdots + t_m\overline{e}_m)$ if every monomial in f has multidegree $(s_1\overline{e}_1 + \cdots + s_n\overline{e_n}) \oplus (t_1\overline{e}_1 + \cdots + t_m\overline{e}_m)$.

The following lemma gives the multidegree of bideterminants. In particular, the multidegree of canonical and anti-canonical bideterminants will allow us to extract terms of a particular shape.

Lemma 3.9: Fix a natural number $n \in \mathbb{Z}_{\geq 0}$. Let $\sigma = (\sigma_1 \geq \cdots \geq \sigma_d)$ be a partition such that the length σ_i of each row is at most n. Let S, T be two conjugate semistandard Young tableaux. Then a bideterminant $(S \mid T)(X)$ is multihomogeneous of multidegree $(s_1\overline{e}_1 + \cdots + s_n\overline{e_n}) \oplus (t_1\overline{e}_1 + \cdots + t_n\overline{e}_n)$, where s_i is equal to the number of i's that appear in S and t_j is equal to the number of j's that appear in T. In particular, $(K_\sigma \mid K_\sigma)(X)$ is multihomogeneous of multidegree $(\widehat{\sigma}_1\overline{e}_1 + \cdots + \widehat{\sigma}_n\overline{e}_1) \oplus (\widehat{\sigma}_1\overline{e}_1 +$

$\cdots + \widehat{\sigma}_n \overline{e}_1)$. Moreover, for distinct partitions $\sigma \neq \tau$, the multidegrees of $(K_\sigma \mid K_\sigma)(X)$ and $(K_\tau \mid K_\tau)(X)$ are distinct, and the same holds for the multidegrees of $(\overline{K}_\sigma \mid \overline{K}_\sigma)(X)$ and $(\overline{K}_\tau \mid \overline{K}_\tau)(X)$.

Proof: The multidegree of a standard bideterminant $(S \mid T)(X)$ is immediate from Definition 3.1. To see that for distinct partitions $\sigma \neq \tau$ the multidegrees of $(K_\sigma \mid K_\sigma)(X)$ and $(K_\tau \mid K_\tau)(X)$ are distinct, we make the following observation. The number of 1's in K_σ is the coefficient of $\overline{e}_1$ in the multidegree, the number of 2's in K_σ is the coefficient of $\overline{e}_2$, and so on. This allows us to get the values $\widehat{\sigma}_1, \ldots, \widehat{\sigma}_n$ which in turn completely determines σ. □

3.2. **Pfaffian Ideals.** We now introduce notation for Pfaffian ideals. Let $\mathbb{F}$ be a field. Let $x_{i,j}$ be indeterminates for $1 \leq i < j \leq 2n$ and let $x_{j,i} := -x_{i,j}$ for $1 \leq j < i \leq 2n$ and $x_{i,i} = 0$ for $1 \leq i \leq 2n$. Thus, $X = (x_{i,j})_{1 \leq i,j \leq 2n}$ is a skew-symmetric $2n \times 2n$ matrix of variables. For notational consistency with the prior subsections, we denote the polynomial ring $\mathbb{F}[x_{i,j} \mid 1 \leq i < j \leq 2n]$ by $\mathbb{F}[X]$.

Recall that for a skew-symmetric matrix X, the determinant when viewed as a polynomial in the entries of the skew-symmetric matrix is a perfect square of some other polynomial. The square root of $\det_{2n}(X)$ is called the *Pfaffian* of X. For the Pfaffian of a $2n \times 2n$ matrix $M \in \mathbb{F}^{2n \times 2n}$, we use the notation pfaff_{2n}. The Pfaffian has a more intrinsic definition given as follows:

$$\mathrm{pfaff}_{2n}(X) = \frac{1}{2^n n!} \sum_{\sigma \in \mathfrak{S}_{2n}} \mathrm{sgn}(\sigma) \prod_{i=1}^{n} x_{\sigma(2i-1),\sigma(2i)}. \tag{2}$$

Just like the determinant, this is a polynomial with coefficients ± 1 so that it is well defined over any field. To see this, note that every monomial in Equation (2) appears exactly $2^n n!$ times. We also have that if m is odd, then $\det_m(X) = 0$ when X is a $m \times m$ skew-symmetric matrix so that $\mathrm{pfaff}_m(X) = 0$. Thus, we only consider Pfaffians on matrices of even order.

Pfaffians are deeply connected to determinants, as indicated by the following lemma:

Lemma 3.10 ([AF22, Lemma 2.27]): Let X be a $2n \times 2n$ skew-symmetric matrix and let E be an arbitrary $2n \times 2n$ matrix. Then $EXE^\top$ is also skew-symmetric and $\mathrm{pfaff}_{2n}(EXE^\top) = \det_{2n}(E)\,\mathrm{pfaff}_{2n}(X)$.

Since the Pfaffian is only well-defined for skew-symmetric matrices, we cannot consider all possible row and column combinations when looking at Pfaffians defined on minors of a matrix.

Definition 3.11 (Principal bitableau, bipfaffian): A bitableau $(S \mid T)$ is *principal* if $S = T$, i.e., if the submatrix of X with rows given by S and columns given by T is principal. In this case, we denote $[T] := (T \mid T)$. Since $m \times m$ determinants of skew-symmetric matrices are zero for odd m, we will assume all partitions have only even row lengths in this case. In particular, $|\sigma|$ will always be divisible by 2.

Let σ be a partition such that the length of each row of σ is even, and let T be a Young tableau of shape σ. Writing the rows as $T_i = (T(i,1), \ldots, T(i,\sigma_i))$, the submatrix of X with rows and columns given by T_i is skew-symmetric as well and thus has a well-defined *principal pfaffian*. The associated *bipfaffian*[†] $[T](X)$ is the product of all such Pfaffians defined by the rows of T:

$$[T](X) := \prod_{i=1}^{\widehat{\sigma}_1} \mathrm{pfaff}\left(x_{T(i,a),T(i,b)}\right)_{1\leq a,b\leq \sigma_i}.$$

Note that the total degree $\deg([T](X))$ is equal to $|\sigma|/2$.

The following lemma shows that every monomial in $\mathbb{F}[X]$ can be expressed as a bipfaffian and hence the bipfaffians span $\mathbb{F}[X]$. The proof is immediate from the definition.

Lemma 3.12: A degree d monomial $\prod_{i=1}^{d} x_{r_i,c_i}$ where for each $1 \leq i \leq d$, $r_i < c_i$ is given by a bipfaffian:

$$\left[\begin{array}{|c|c|}\hline r_1 & c_1 \\ \hline r_2 & c_2 \\ \hline \vdots & \vdots \\ \hline r_d & c_d \\ \hline \end{array}\right](X) = \prod_{i=1}^{d} x_{r_i,c_i}.$$

Analogous to the classical determinantal case, a specific subset of bipfaffians form an $\mathbb{F}$-basis of $\mathbb{F}[X]$.

Definition 3.13 (Standard bipfaffian): A bipfaffian $[S](X)$ is *standard* if S is a conjugate semistandard Young tableau.

As in the determinantal case, the Pfaffian case also admits a straightening law.

Theorem 3.14 ([dCP76, Theorem 6.5], [HT92, §5]): Let S be a Young tableau of shape σ. Then $[S](X)$ can be uniquely expressed as a linear combination

$$[S](X) = \sum_{A} c_A [A](X),$$

where $[A](X)$ is a standard bipfaffian of shape $\tau \geq_{\mathrm{lex}} \sigma$, and $c_A \in \mathbb{Z}$ when $\mathrm{char}(\mathbb{F}) = 0$, while $c_A \in \mathbb{Z}/p\mathbb{Z}$ when $\mathrm{char}(\mathbb{F}) = p > 0$.

Lemma 3.12 and Theorems 3.14 together imply the following corollary.

Corollary 3.15: The standard bipfaffians form an $\mathbb{F}$-basis of $\mathbb{F}[X]$. In particular, since bipfaffians are homogeneous with respect to total degree, the degree-d component of $\mathbb{F}[X]$ has as basis the standard bipfaffians of shape σ such that $|\sigma| = 2d$. Note that if $|\sigma| = 2d$, then the number of rows $\widehat{\sigma}_1$ of σ is at most d as each non-empty row of σ has length at least 2.

[†] [AF22] call this a *standard monomial*. We deviate from this as standard monomial is a more general term which also refers to standard bideterminants and other analogues in standard monomial theory.

Let $I^{\mathrm{pfaff}}_{2n,2r}$ be the ideal of $\mathbb{F}[X]$ generated by the $2r \times 2r$ principal bipfaffians of X. We have the following crucial statement.

Proposition 3.16 ([HT92, §5]): Polynomials $f \in I^{\mathrm{pfaff}}_{2n,2r}$ are supported by standard bipfaffians of shape σ such that $\sigma_1 \geq 2r$.

Next, we describe a grading of the polynomial ring $\mathbb{F}[X]$ that is useful for the proof.

Definition 3.17: The polynomial ring $\mathbb{F}[X]$ has a natural $\mathbb{N}^{2n}$-grading. Let $\overline{e}_i \in \mathbb{N}^{2n}$ be the standard basis vector with a 1 in position i and 0's in the other positions. Then we assign $x_{i,j}$ to have degree $\overline{e}_i + \overline{e}_j$. We call this assignment the *multidegree*. We say a polynomial $f \in \mathbb{F}[X]$ is *multihomogeneous* of multidegree $s_1\overline{e}_1 + \cdots + s_{2n}\overline{e_{2n}}$ if every monomial in f has multidegree $s_1\overline{e}_1 + \cdots + s_{2n}\overline{e_{2n}}$.

The following lemma gives the multidegree of bipfaffians. It follows immediately from the definition and is proven similarly to Lemma 3.9 and thus we omit the proof.

Lemma 3.18: Fix a natural number $n \in \mathbb{Z}_{\geq 0}$. Let $\sigma = (\sigma_1 \geq \cdots \geq \sigma_k)$ be a partition such that the length of each row is at most $2n$. Let S be a conjugate semistandard Young tableau. Then a bipfaffian $[S](X)$ is multihomogeneous of multidegree $s_1\overline{e}_1 + \cdots + s_{2n}\overline{e}_{2n}$, where s_i is equal to the number of i's that appear in S. In particular, $[K_\sigma](X)$ is multihomogeneous of multidegree $\widehat{\sigma}_1\overline{e}_1 + \cdots + \widehat{\sigma}_{2n}\overline{e}_{2n}$. Moreover, for distinct partitions $\sigma \neq \tau$, the multidegrees of $[K_\sigma](X)$ and $[K_\tau](X)$ are distinct, and the same holds for the multidegrees of $[\overline{K}_\sigma](X)$ and $[\overline{K}_\tau](X)$.

3.3. Hankel Ideals.

3.4. Symmetric Determinantal Ideals. ⟨⟨ Only if time ⟩⟩

3.5. Permanental Ideals. ⟨⟨ Only if time ⟩⟩

4. Prior Work

4.1. Generalities on Complexity in Ideals. Roughly speaking, the complexity of an ideal I under a given algebraic model (such as algebraic circuits or algebraic branching programs) is defined as the minimum complexity, in that model, of any nonzero polynomial $f \in I$ [Gro20]. Proving lower bounds for ideals is harder than for individual polynomials, since one must establish a bound for every nonzero $f \in I$ rather than just a single polynomial. This motivates the search for a property of the following form: if an arbitrary nonzero $f \in I$ is efficiently computable in a given model, then some polynomial from a distinguished set $S = \{g_1, \ldots, g_k\}$ (often a natural generating set of I or some related set with desirable properties) is also efficiently computable in the given model. In this way, proving lower bounds for arbitrary nonzero $f \in I$ reduces to proving them for $g_1, \ldots, g_k$. We call such a property a *closure result*, by analogy with closure under factorization in the principal ideal case. In short, closure results reduce the task of showing hardness for all polynomials in I to proving hardness for a small set of distinguished polynomials.

Definition 4.1 (Hitting Set Generator (HSG)):

Definition 4.2 (Polynomial Identity Testing (PIT)):

Closure results for ideals are useful for constructing hitting set generators, which are the algebraic-complexity analogs of pseudorandom generators and can be used to derandomize polynomial identity testing (PIT). The idea is as follows: for $s < n$, we seek a hitting set generator $G\colon \mathbb{F}^s \to \mathbb{F}^n$ against a family $\mathcal{C}$ of n-variate polynomials. This means that G has the property that for any nonzero polynomial $f \in \mathcal{C}$, we have $f \circ G \neq 0$. Here, one should think of $\mathcal{C}$ as a family of "low-complexity" polynomials. Let I be the ideal of polynomials f satisfying $f \circ G = 0$, also known as the *vanishing ideal* of G [HMM24]. To show that G has the desired hitting set property, it suffices to show that every nonzero $f \in I$ has sufficiently high complexity so that no such f can belong to $\mathcal{C}$. With a closure result for I, this reduces to proving lower bounds for a distinguished set of polynomials $\{g_1, \ldots, g_k\}$ for I. This can be viewed as a particular way of implementing the "hardness versus randomness" paradigm of [NW94; KI04] in the setting of algebraic complexity.

The general question we want to study is of the following form:

Question 4.3: Say that $I = \langle g_1, \ldots, g_k \rangle$. Let $0 \neq f \in I$. How does the complexity of f compare to the complexity of the generators $g_1, \ldots, g_k$.

4.1.1. *Principal Ideals.* Let us consider the special case of Question 4.3 where $I = \langle g \rangle$ is a *principal ideal*. Then our question is now the following:

Question 4.4: Let $0 \neq f \in \langle g \rangle$. Does g have a small f-oracle circuit?

The affirmative was actually conjectured by Bürgisser:

Conjecture 4.5 ([Bür00, Conjecture 8.3]): If g is a factor of $f \neq 0$, then $\mathrm{size}(g) \le \mathrm{poly}(\mathrm{size}(f), \deg(g))$.

There has been much progress on Conjecture 4.5. Factoring in algebraic complexity is well studied, see the survey of Bhargav, Dwivedi, and Saxena for an overview of recent progress [BDS26]. For example, it is known that for fields of characteristic zero that $\mathrm{size}(g) \le \mathrm{poly}(\mathrm{size}(f), \deg(f))$ [Kal87; Bür00]. Compare the use of $\deg(f)$ here to $\deg(g)$ in Conjecture 4.5. Thus, the most interesting regime to study here is the situation where the degree of the factor g is significantly smaller than the degree of the whole polynomial f. This result is only known in the setting of border computation over fields of characteristic zero:

Theorem 4.6 ([Bür04, Theorem 1.3]): Let $\mathbb{F}$ be a field of characteristic zero. Let $f(\overline{x}) \neq 0$ be an n-variate polynomial. If $g(\overline{x})$ is a factor of $f(\overline{x})$, then $\overline{\mathrm{size}}(g) \le \mathrm{poly}(\mathrm{size}(f), \deg(g), n)$.

Bürgisser's result can actually be formulated in the setting of oracle complexity, so that we get a poly(deg(g), n) size f-oracle circuit which approximately computes $g(\overline{x})$.

⟨⟨ TODO: Read and understand Kayal's work on annihilator polynomials and how it connects to HSGs ⟩⟩

4.1.2. *PIT vs Factoring.*

4.1.3. *More Than One Generator.*

5. Current Work on Ideals with Straightening Laws

⟨⟨ TODO: motivate Grochow's conjecture ⟩⟩

Conjecture 5.1 ([Gro20, Conjecture 6.3]): Let X be an $n \times n$ matrix of indeterminates and let I_n be the ideal generated by the $\frac{n}{2} \times \frac{n}{2}$ minors of X. Then for every nonzero polynomial $f \in I_n$, there is a constant-depth[‡] algebraic circuit of size poly(n) with f-oracle gates that computes the $m \times m$ determinant for some $m = n^{\Theta(1)}$.

5.1. **Approximative Results.** Analogous to Bürgisser's theorem [Bür04] that the Factor Conjecture holds with respect to border complexity, Andrews and Forbes [AF22] proved that Grochow's conjecture holds with respect to border complexity. Formally, they proved the following theorem:

Theorem 5.2 ([AF22, Theorem 1.1]): Let $\mathbb{F}$ be a field of characteristic zero. Let $X = (x_{i,j})_{\substack{1\le i\le n\\ 1\le j\le m}}$ be an $n \times m$ matrix of variables over $\mathbb{F}$ and let $I^{\det}_{n,m,r} \subseteq \mathbb{F}[X] = \mathbb{F}[x_{i,j}]$ be the ideal generated by the $r \times r$ minors of X. Let $f \in I^{\det}_{n,m,r}$ be a nonzero polynomial. Then there is a depth-three f-oracle circuit of size $O\left(n^2m^2\right)$ that approximately computes the $t \times t$ determinant for $t = \Theta\left(r^{1/3}\right)$.

Theorem 5.3 ([AF22, Theorem 1.1]): Let $\mathbb{F}$ be a field of characteristic zero. Let $X = (x_{i,j})_{1\le i,j\le 2n}$ be a $2n \times 2n$ matrix of variables over $\mathbb{F}$ and let $I^{\text{pfaff}}_{2n,2r} \subseteq \mathbb{F}[X] = \mathbb{F}[x_{i,j}]$ be the ideal generated by the Pfaffians of the $2r \times 2r$ principal submatrices of X. Let $f \in I^{\text{pfaff}}_{2n,2r}$ be a nonzero polynomial. Then there is a depth-three f-oracle circuit of size $O\left(n^4\right)$ that approximately computes the $t \times t$ Pfaffian for $t = \Theta\left(r^{1/3}\right)$.

5.2. **Debordering.** In [DG25], we deborder the results of [AF22]. This is done via a novel application of the Isolation Lemma: Lemma 2.25 and Corollary 2.26.

Theorem 5.4 (Informal version of [DG25, Theorem 3.18, Corollary 3.19]): Let $X = (x_{i,j})_{\substack{1\le i\le n\\ 1\le j\le m}}$ be an $n \times m$ matrix of variables over a field $\mathbb{F}$. Let $I^{\det}_{n,m,r} \subseteq \mathbb{F}[X] = \mathbb{F}[x_{i,j} \mid 1 \le i \le n,\ 1 \le j \le m]$ be the ideal generated by the $r \times r$ minors of X. Let $f \in I^{\det}_{n,m,r}$ be a nonzero polynomial. Assume that the characteristic of $\mathbb{F}$ is either zero or greater than deg(f). Also assume that the size of $\mathbb{F}$ is sufficiently

[‡]The original statement of the conjecture does not specify constant-depth algebraic circuits. However, it is already known that the $m \times m$ determinant can be computed by algebraic circuits of size poly(m).

large.[§] Then there exists a depth-three f-oracle circuit of size $\mathrm{poly}(n, m, \deg(f))$ that computes the $t \times t$ determinant for $t = \Theta\left(r^{1/3}\right)$.

Theorem 5.5 (Informal version of [DG25, Theorem 4.10, Corollary 4.11]): Let $X = (x_{i,j})_{1 \le i < j \le 2n}$ be a $2n \times 2n$ skew-symmetric matrix of variables over a field $\mathbb{F}$ so that $x_{j,i} = -x_{i,j}$ for $1 \le j < i \le n$ and $x_{i,i} = 0$ for $1 \le i \le 2n$. Let $I^{\text{pfaff}}_{2n,2r} \subseteq \mathbb{F}[X] = \mathbb{F}[x_{i,j} \mid 1 \le i < j \le 2n]$ be the ideal generated by the $2r \times 2r$ principal Pfaffians of X. Let $f \in I^{\text{pfaff}}_{2n,2r}$ be a nonzero polynomial. Assume that the characteristic of $\mathbb{F}$ is either zero or greater than $\deg(f)$. Also assume that the size of $\mathbb{F}$ is sufficiently large. Then there exists a depth-three f-oracle circuit of size $\mathrm{poly}(n, \deg(f))$ that computes the $t \times t$ Pfaffian of a skew-symmetric matrix for $t = \Theta\left(r^{1/3}\right)$.

5.3. Proof Overview for Determinantal Case.

5.3.1. *Linear Operators.* Our application of the determinantal straightening law (Corollary 3.6), together with the isolation lemma (Corollary 2.26) and extraction of coefficients (Lemma 2.23), allows us to isolate a canonical bitableau from the expansion of nonzero $f \in I^{\det}_{n,m,r}$ in the basis of standard bideterminants. In particular, we will show that this can be done via a polynomial-sized depth-three f-oracle circuit using the isolation lemma (Corollary 2.26) as well as coefficient extraction (Lemma 2.23). This is in contrast to the proof of [AF22], where they use a Kronecker-type substitution in order to isolate terms. To do this, we will give a more careful analysis of the terms that arise when applying the substitution operators to the expression of a polynomial $f(X) \in I^{\det}_{n,m,r}$.

Throughout, fix a field $\mathbb{F}$. Let n, m be natural numbers, and for $1 \le i \le n$ and $1 \le j \le m$, let $x_{i,j}$ be indeterminates. Throughout this section, let $X = (x_{i,j})$ be an $n \times m$ matrix of variables. Let $1 \le r \le \min(n, m)$ and define $I^{\det}_{n,m,r}$ to be the ideal in $\mathbb{F}[X]$ generated by the $r \times r$ minors of X.

Our first step will be to establish a series of linear transformations that send a conjugate semistandard bitableau to the anti-canonical conjugate semistandard bitableau of the same shape.

Definition 5.6 (Substitution operator): Let $i < j$. Define the operator $\mathrm{Sub}_{i \to j}$ which takes as input a tableau S and returns the tableau S' which is formed by taking each row of S which has an i but not a j, replacing that i with j, and then sorting the row in increasing order. We also define $h_i^j(S)$ to be the number of times i is replaced by j after applying $\mathrm{Sub}_{i \to j}$ to S.

Definition 5.7 (Canonical/anti-canonical tableau): Fix a natural number n. Let $\sigma = (\sigma_1 \ge \cdots \ge \sigma_k > 0)$ be a partition such that the length σ_i of each row is at most n. Define $K_{\sigma,n}$ to be the tableau of shape σ where row i has entries $1, 2, \ldots, \sigma_i$. Similarly, define $\overline{K}_{\sigma,n}$ to be the tableau of shape σ where row i has entries $(n - \sigma_i + 1, n - \sigma_i + 2, \ldots, n)$. We call $K_{\sigma,n}$ and $\overline{K}_{\sigma,n}$ the *canonical* and *anti-canonical*

[§]Specifically, we require $|\mathbb{F}|$ to be bounded below by some polynomial in n, m, and $\deg(f)$. Such assumptions are standard in the literature (for instance, in work on polynomial factorization), since one can often achieve them by passing to a suitable field extension.

tableau respectively.[¶] Note that $K_{\sigma,n}$ and $\overline{K}_{\sigma,n}$ are both conjugate semistandard. When n is clear from context, we will just write K_σ and $\overline{K}_\sigma$.

Example 5.8: If $\sigma = (5,4,2,1)$ and $n = 7$, then K_σ and $\overline{K}_\sigma$ are given in Figure 1.

$$\begin{array}{|c|c|c|c|c|}\hline 1 & 2 & 3 & 4 & 5 \\ \hline 1 & 2 & 3 & 4 \\ \cline{1-4} 1 & 2 \\ \cline{1-2} 1 \\ \cline{1-1} \end{array} \qquad \begin{array}{|c|c|c|c|c|}\hline 3 & 4 & 5 & 6 & 7 \\ \hline 4 & 5 & 6 & 7 \\ \cline{1-4} 6 & 7 \\ \cline{1-2} 7 \\ \cline{1-1} \end{array}$$

FIGURE 1. The canonical and anti-canonical tableau for $\sigma = (5,4,2,1)$ and $n = 7$.

We now study how these substitution operators act on bideterminants $(S \mid T)(X)$. From properties of the determinant, we see that multiplying the matrix of variables X by a elementary matrix $E_{i,j}$ corresponds to applying substitution operators on S or T:

Lemma 5.9: Let λ be a new indeterminate. Let $(S \mid T)$ be a bitableau, not necessarily standard, of shape σ. For $0 \le h \le h_i^j(S) - 1$, let $\mathcal{C}^h_{i\to j}(S)$ be the set of tableaux of shape σ obtained by changing i to j at exactly h rows of S which contain i but not j and reordering those rows to be increasing. Then we have that

$$(S \mid T)(E_{i,j}(\lambda)X) = \pm\lambda^{h_i^j(S)}(\mathrm{Sub}_{i\to j}(S) \mid T)(X) + \sum_{h=0}^{h_i^j(S)-1} \lambda^h \sum_{S' \in \mathcal{C}^h_{i\to j}(S)} \pm(S' \mid T)(X). \tag{3}$$

Similarly, we also have that

$$(S \mid T)(XE_{i,j}(\lambda)^\top) = \pm\lambda^{h_i^j(T)}(S \mid \mathrm{Sub}_{i\to j}(T))(X) + \sum_{h=0}^{h_i^j(T)-1} \lambda^h \sum_{T' \in \mathcal{C}^h_{i\to j}(T)} \pm(S \mid T')(X). \tag{4}$$

Definition 5.10: In addition to the variables $x_{i,j}$, where $i \in [n]$ and $j \in [m]$, we introduce new sets of indeterminates $\Lambda = \{\lambda_{i,j} \mid 1 \le i < j \le n\}$ and $\Xi = \{\xi_{i,j} \mid 1 \le i < j \le m\}$. We have that $|\Lambda| = \binom{n}{2} = O\left(n^2\right)$ and $|\Xi| = \binom{m}{2} = O\left(m^2\right)$. The purpose of these new indeterminates is that they will allow us to keep track of the effects of substitution operators as we isolate a canonical bitableau.

Define $M \in \mathbb{F}[\Lambda]^{n\times n}$ and $N \in \mathbb{F}[\Xi]^{m\times m}$ by

$$\begin{aligned} M = {} & E_{1,2}(\lambda_{1,2})\cdots E_{1,n}(\lambda_{1,n})E_{2,3}(\lambda_{2,3})\cdots E_{2,n}(\lambda_{2,n}) \\ & \cdots E_{n-2,n-1}(\lambda_{n-2,n-1})E_{n-2,n}(\lambda_{n-2,n})E_{n-1,n}(\lambda_{n-1,n}) \end{aligned}$$

[¶]In some sources, such as [BV88], these are referred to as the *initial* and *final* tableau.

and

$$N = E_{m-1,m}(\xi_{m-1,m})^\top E_{m-2,m}(\xi_{m-2,m})^\top E_{m-2,m-1}(\xi_{m-2,m-1})^\top \cdots E_{2,m}(\xi_{2,m})^\top \cdots E_{2,3}(\xi_{2,3})^\top E_{1,m}(\xi_{1,m})^\top \cdots E_{1,2}(\xi_{1,2})^\top.$$

For an integer $k > 0$, let $J_k \in \mathbb{F}^{k\times k}$ be the $k \times k$ matrix with 1's along the anti-diagonal from the bottom left to the top right and 0's elsewhere. Note that $\det_k(J_k) = (-1)^{\binom{k}{2}} = \pm 1$. The substitution $X \mapsto J_n X J_m$ has the effect of replacing row i with row $n-i+1$ and column j with column $m-j+1$. This turns an anti-canonical bitableau $(\overline{K}_\sigma \mid \overline{K}_\sigma)$ into a canonical one $(K_\sigma \mid K_\sigma)$.

5.3.2. *Isolation of a Single Term.*

Corollary 5.11: Let $f(X) \in I^{\det}_{n,m,r}$ be a nonzero polynomial of degree d. Then $f(MJ_nXJ_mN)$ can be expressed as a sum

$$f(MJ_nXJ_mN) = \sum_{k\in A} \widehat{c}_k \Lambda^{\overline{e}_k} \Xi^{\overline{f}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X) + \sum_{\ell\in B} \widehat{c}_\ell \Lambda^{\overline{e}_\ell} \Xi^{\overline{f}_\ell} \cdot (S_\ell \mid T_\ell)(X), \tag{5}$$

such that the following hold:

(1) A is a nonempty finite set and B is finite set disjoint from A.

(2) All terms in Equation (5) have the form $c\Lambda^{\overline{e}}\Xi^{\overline{f}} \cdot (S \mid T)(X)$, where $c \in \mathbb{F}^\times$, $\overline{e} \in \{0,1,\dots,d\}^{\binom{[n]}{2}}$, $\overline{f} \in \{0,1,\dots,d\}^{\binom{[m]}{2}}$, and $(S \mid T)(X)$ is a bideterminant in $I^{\det}_{n,m,r}$ of degree d. In particular, if S and T have shape σ, then $|\sigma| \le d$.

(3) The triples $(\overline{e}_k, \overline{f}_k, \sigma_k)$ are distinct, where k ranges over A.

Next, we introduce i variables $Y = \{y_1,\dots,y_n\}$ and $Z = \{z_1,\dots,z_m\}$. Define the diagonal matrices

$$D = \operatorname{diag}(y_1,\dots,y_n) \in \mathbb{F}[Y]^{n\times n} \quad \text{and} \quad D' = \operatorname{diag}(z_1,\dots,z_m) \in \mathbb{F}[Z]^{m\times m}.$$

Lemma 5.12: Let $(S \mid T)(X)$ be a bideterminant of degree at most d. Then

$$(S \mid T)(DXD') = Y^{\overline{s}} Z^{\overline{t}} \cdot (S \mid T)(X),$$

where $\overline{s} = (s_1,\dots,s_n) \in \{0,1,\dots,d\}^n$, $\overline{t} = (t_1,\dots,t_m) \in \{0,1,\dots,d\}^m$, s_i is the number of times i appears in S, t_j is the number of times j appears in T, i.e., $(s_1e_1 + \cdots + s_ne_n) \oplus (t_1e_1 + \cdots + t_me_m)$ is the multidegree of $(S \mid T)(X)$ with respect to the grading defined in Definition 3.8.

We will often simply say that $(S \mid T)(X)$ in Lemma 5.12 has multidegree $(\overline{s}, \overline{t})$ rather than writing it as $(s_1e_1 + \cdots + s_ne_n) \oplus (t_1e_1 + \cdots + t_me_m)$.

Next, we introduce another variable v, and modify D and D' as follows:

$$\widetilde{D} = \mathrm{diag}(y_1 v, y_2 v^2 \dots, y_n v^n) \in \mathbb{F}[Y, v]^{n\times n} \quad \text{and} \quad \widetilde{D}' = \mathrm{diag}(z_1 v, z_2 v^2, \dots, z_m v^m) \in \mathbb{F}[Z, v]^{m\times m}.$$

Lemma 5.13: Let $(S \mid T)(X)$ be a bideterminant. Then

$$(S \mid T)(\widetilde{D}X\widetilde{D}') = Y^{\bar{s}} Z^{\bar{t}} v^{|S|+|T|} \cdot (S \mid T)(X),$$

where $(\bar{s}, \bar{t})$ is the multidegree of $(S \mid T)(X)$.

Lemma 5.14: Let $f(X) \in I^{\det}_{n,m,r}$ be a nonzero polynomial of degree d. Let $g = f(MJ_n\widetilde{D}X\widetilde{D}'J_mN) \in \mathbb{F}[X, \Lambda, \Xi, Y, Z, v]$. View g as a univariate polynomial in v with coefficients in $\mathbb{F}[X, \Lambda, \Xi, Y, Z]$, and write $g = \sum_i \mathrm{coeff}_{v^i}(g)v^i$, where $\mathrm{coeff}_{v^i}(g) \in \mathbb{F}[X, \Lambda, \Xi, Y, Z]$ denotes the coefficient of v^i in g. Choose the smallest integer $d_{\min}$ such that $\mathrm{coeff}_{v^{d_{\min}}}(g) \neq 0$. Then $d_{\min} = 2|K_\sigma|$ for some shape σ with $|\sigma| \le d$ and we may write $\mathrm{coeff}_{v^{d_{\min}}}(g)$ as a finite sum

$$\mathrm{coeff}_{v^{d_{\min}}}(g) = \sum_{k\in I} c_k \Lambda^{\bar{e}_k} \Xi^{\bar{f}_k} Y^{\bar{s}_k} Z^{\bar{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X), \tag{6}$$

such that the following hold:

(1) For each $k \in I$, $c_k \in \mathbb{F}^\times$, $(\sigma_k)_1 \ge r$, and $|\sigma_k| \le d$.

(2) The tuples $(\bar{e}_k, \bar{f}_k, \bar{s}_k, \bar{t}_k)$ are distinct, where k ranges over I.

Lemma 5.14 helps us separate a collection of terms solely containing canonical bideterminants. The next lemma further applies the isolation lemma to single out one term from this collection.

Lemma 5.15: Let $f(X) \in I^{\det}_{n,m,r}$ be a nonzero polynomial of degree d. Let $g = f(MJ_n\widetilde{D}X\widetilde{D}'J_mN) \in \mathbb{F}[X, \Lambda, \Xi, Y, Z, v]$. Then there exist integers $z_t = O\big(d(n^2+m^2)\big)$ for each variable $t \in \Lambda \sqcup \Xi \sqcup Y \sqcup Z$, and an integer $z_v = O\big(d^2(n^2+m^2)\big)$ such that for the two variable substitution maps

$$\phi : t \mapsto w^{z_t} \quad \text{for } t \in \Lambda \sqcup \Xi \sqcup Y \sqcup Z$$

and

$$\psi : v \mapsto w^{z_v},$$

we have that $h := (\psi \circ \phi)(g) \in \mathbb{F}[X, w]$ has $\deg_w(h) = O\big(d^3(n^3+m^3)\big)$, and that $\mathrm{coeff}_{w^i}(h) = c \cdot (K_\sigma \mid K_\sigma)(X)$ for some integer $i \le \deg_w(h)$, where $c \in \mathbb{F}^\times$, $\sigma_1 \ge r$ and $|\sigma| \le d$.

Lemma 5.14 helps us separate a collection of terms solely containing canonical bideterminants. The next lemma further applies the isolation lemma to single out one term from this collection.

Theorem 5.16: Let $f(X) \in I^{\det}_{n,m,r}$ be a nonzero polynomial of degree d. Assume $|\mathbb{F}| \ge c_0 d^3(n^3 + m^3)$, where $c_0 > 0$ is a large enough constant. Then, there exists a depth-three f-oracle circuit of size $O\left(n^2m^2d^3(n^3+m^3)\right) = \mathrm{poly}(n,m,d)$ computing $(K_\sigma \mid K_\sigma)(X)$, where $\sigma_1 \geq r$ and $|\sigma| \leq d$. Furthermore, the top gate of this circuit is an addition gate, and the bottom layer consists of $O\left(nmd^3(n^3+m^3)\right)$ addition gates. The total number of gates and wires, excluding those between the bottom addition gates and the input gates, is $O\left(nmd^3(n^3+m^3)\right)$.

5.3.3. *Projection to Determinant.* We have now seen that given an oracle to some nonzero $f(X) \in I^{\det}_{n,m,r}$, one can efficiently compute a canonical bideterminant efficiently with an f-oracle circuit. We now leverage the connection between algebraic branching programs The remainder of the proof follows [AF22], which we include here for completeness. One key idea is exploiting the close connection between algebraic branching programs and determinants (Lemma 2.11 and Lemma 2.12).

We will use Theorems 5.16 to construct a constant-depth algebraic circuit of size $\mathrm{poly}(n, \deg(f))$ with f-oracle gates that computes the determinant. We start by proving the following more general theorem.

Theorem 5.17: Let $f(X) \in I^{\det}_{n,m,r}$ be a nonzero polynomial of degree d. Let $g(y_1,\ldots,y_\ell) \in \mathbb{F}[\overline{y}]$ be a polynomial computable by an algebraic branching program with at most r vertices. Assume $|\mathbb{F}| \geq c_0 d^3(n^3+m^3)$, where $c_0 > 0$ is a large enough constant. Then there is a depth-three f-oracle circuit Φ of size $O\left(nmd^4\ell r(n^3+m^3)\right)$ defined over $\mathbb{F}$ such that

- if $\mathrm{char}(\mathbb{F}) = 0$ then Φ computes $g(\overline{y})$, and
- if $\mathrm{char}(\mathbb{F}) = p > 0$ then Φ computes $g(\overline{y})^{p^k}$ for some $k \leq \lfloor \log_p(d) \rfloor$.

Furthermore, the top layer of this circuit consists of an addition gate.

Proof: By Lemma 2.12, there is a homogeneous polynomial $\widehat{g}(\overline{y},z) \in \mathbb{F}[\overline{y},z]$ such that $\widehat{g}(\overline{y},1) = g(\overline{y})$ and $\widehat{g}(\overline{y},z)$ can be computed by an algebraic branching program over $\mathbb{F}$ with at most r vertices. We may add isolated vertices such that the algebraic branching program has exactly r vertices. Then by Lemma 2.11, there is a matrix $A(\overline{y},z) \in \mathbb{F}[\overline{y},z]^{r\times r}$ such that

- $\det_r(A(\overline{y},z)) = 1 + \widehat{g}(\overline{y},z)$,
- $\det_i(A(\overline{y},z)_{[i],[i]}) = 1$ for all $1 \leq i \leq r-1$,

and all entries of $A(\overline{y},z)$ have degree at most 1 in $\overline{y}$ and z. Extend $A(\overline{y},z)$ to an $n\times m$ matrix by adding 1's to the main diagonal and 0's elsewhere.

By Theorems 5.16, there exists a depth-three f-oracle circuit C over $\mathbb{F}$ of size $O\left(n^2m^2d^3(n^3+m^3)\right)$ computing $(K_\sigma \mid K_\sigma)(X)$ such that $\sigma_1 \geq r$ and $|\sigma| \leq d$. Furthermore, the top gate of C is an addition gate, the bottom layer of C consists of $O\left(nmd^3(n^3+m^3)\right)$ addition gates, and the total number of gates and wires excluding the wires between the bottom addition gates and the input gates is $O\left(nmd^3(n^3+m^3)\right)$. Replacing the nm input gates of C by the $O(\ell)$ new input gates $\overline{y}$, z, and 1, and further connecting these gates to the $O\left(nmd^3(n^3+m^3)\right)$ bottom addition gates, we obtain a depth-three f-oracle circuit over $\mathbb{F}$ of size $O\left(nmd^3\ell(n^3+m^3)\right)$ computing

$$h(\overline{y},z) := (K_\sigma \mid K_\sigma)(A(\overline{y},z)) = (1+\widehat{g}(\overline{y},z))^t$$

where $t := |\{i \mid \sigma_i \geq r\}| \geq 1$. Note that $t \leq \widehat{\sigma}_1 \leq d$.

First, suppose that $\mathrm{char}(\mathbb{F}) = 0$. Let δ be a new indeterminate. Then under the map $y_i \mapsto \delta \cdot y_i$ and $z \mapsto \delta$, we have that

$$\begin{aligned} h(\delta\cdot\overline{y},\delta) &= (1+\widehat{g}(\delta\cdot\overline{y},\delta))^t \\ &= (1+\delta^{\deg(\widehat{g})}\widehat{g}(\overline{y},1))^t \\ &= (1+\delta^{\deg(\widehat{g})}g(\overline{y}))^t \\ &= \sum_{i=0}^{t}\binom{t}{i}\delta^{i\cdot\deg(\widehat{g})}g(\overline{y})^i = 1+\delta^{\deg(\widehat{g})}g(\overline{y})+\cdots. \end{aligned} \tag{7}$$

Then $\deg_\delta(h(\delta\cdot\overline{y},\delta)) \leq t\cdot\deg(\widehat{g}) \leq dr$. For each $\alpha \in \mathbb{F}$, we can construct from the depth-three f-oracle circuit computing $h(\overline{y},z)$ another f-oracle circuit C_α computing $h(\alpha\overline{y},\alpha)$, whose size is $O\left(nmd^3\ell(n^3+m^3)\right)$ and top gate is an addition gate. Also, as $|\mathbb{F}| \geq c_0 d^3(n^3+m^3)$, where $c_0 > 0$ is a large enough constant, we may assume $|\mathbb{F}| \geq dr+1$. Therefore, by Lemma 2.23, we can compute $\mathrm{coeff}_{\delta^{\deg(\widehat{g})}}(h(\delta\cdot\overline{y},\delta)) = g(\overline{y})$ using a depth-three f-oracle circuit of size

$$O\left(\deg_\delta(h(\delta\cdot\overline{y},\delta))\cdot nmd^3\ell(n^3+m^3)\right) = O\left(nmd^4\ell r(n^3+m^3)\right).$$

Now suppose that $\mathrm{char}(\mathbb{F}) = p > 0$. The above computation only needs to be modified in the case that $p \mid t$. Let $k \in \mathbb{N}$ be the largest natural number such that $t = p^k b$ for some natural number b. Since $p^k \leq t \leq d$, we have that $k \leq \lfloor \log_p(d) \rfloor$ as claimed. Then by a similar computation to Equation (7), we get that

$$h(\delta\cdot\overline{y},\delta) = \sum_{i=0}^{t}\binom{t}{i}\delta^{i\cdot\deg(\widehat{g})}g(\overline{y})^i = 1+b\cdot\delta^{\deg(\widehat{g})p^k}g(\overline{y})^{p^k}+\cdots. \tag{8}$$

Again, $\deg_\delta(h(\delta\cdot\overline{y},\delta)) \leq t\cdot\deg(\widehat{g}) \leq dr$. For each $\alpha \in \mathbb{F}$, we can construct from the depth-three f-oracle circuit computing $h(\overline{y},z)$ another f-oracle circuit C_α computing $h(\alpha\overline{y},\alpha)$, whose size is $O\left(nmd^3\ell(n^3+m^3)\right)$ and top gate is an addition gate. Also, as $|\mathbb{F}| \geq c_0 d^3(n^3+m^3)$, where $c_0 > 0$ is a large enough constant, we may assume $|\mathbb{F}| \geq dr+1$. Therefore, by Lemma 2.23, we have a depth-three f-oracle circuit of size $O\left(nmd^4\ell r(n^3+m^3)\right)$ computing $\mathrm{coeff}_{\delta^{\deg(\widehat{g})p^k}}(h(\delta\cdot\overline{y},\delta)) = b\cdot g(\overline{y})^{p^k}$. By dividing the multipliers on the wires connecting to the top gate by b, this circuit computing $b\cdot g(\overline{y})^{p^k}$ can be turned into a depth-three f-oracle circuit with the same underlying graph that computes $g(\overline{y})^{p^k}$. $\square$

The following corollaries are a debordering of the results [AF22, Corollary 3.9, Corollary 3.10] of Andrews and Forbes. In particular, we can use Theorems 5.17 to construct a small constant-depth f-oracle circuit for the determinant, which partially resolves Conjecture 5.1 posed by Grochow as stated before.

Corollary 5.18: Let $f(X) \in I^{\det}_{n,m,r}$ be a nonzero polynomial of degree d. Let $t = O\left(r^{1/3}\right)$ and let Y be a $t \times t$ matrix of indeterminates. Assume $|\mathbb{F}| \geq c_0 d^3(n^3 + m^3)$, where $c_0 > 0$ is a large enough constant. Then there is a depth-three f-oracle circuit Φ of size $O\left(nmd^4 r t^2(n^3+m^3)\right) \leq O\left(nmd^4 r^{5/3}(n^3+m^3)\right)$ over $\mathbb{F}$ such that

- if $\mathrm{char}(\mathbb{F}) = 0$ then Φ computes $\det_t(Y)$, and
- if $\mathrm{char}(\mathbb{F}) = p > 0$ then Φ computes $\det_t(Y)^{p^k}$ for some $k \leq \lfloor \log_p(d) \rfloor$.

Furthermore, the top gate of this circuit is an addition gate.

Proof: Mahajan and Vinay [MV97, Theorem 2] construct an algebraic branching program on $O\left(t^3\right) \leq r$ vertices which computes $\det_t(Y)$. Then, the total number of variables in Y is $t^2 = O\left(r^{2/3}\right)$. The result then follows by applying Theorems 5.17. □

6. Future Work and Open Problems

6.1. Symmetric Determinantal Ideals.

6.2. Permanental Ideals. ⟨⟨ if time ⟩⟩

6.3. Hankel and Toeplitz Ideals.

6.4. Complexity of Straightening. ⟨⟨ if time ⟩⟩

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